THE MODULE OF DIFFERENTIALS OF A SCHEME OVER ITS BASE EQUALS THE MODULE OF DIFFERENTIALS OVER ITS RELATIVE FROBENIUS FACTOR

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof cites no external \ref inside the counted span, so the assembly needs no external-reference stubs.

Allowed standard prerequisites (editorial).

- Module of Kaehler differentials and the universal derivation. Stacks Project, Algebra (module of differentials and base change). The whole proof runs on the universal property of Omega_{B/A} and its behaviour under the Frobenius-linear base change.

Lemma 0.1. *Let $p > 0$ be a prime number. Let S be a scheme in characteristic p. Let X be a scheme over S. Then $\Omega_{X/S} = \Omega_{X/X^{(p)}}$.*

Proof. This translates into the following algebra fact. Let $A \to B$ be a homomorphism of rings of characteristic p. Set $B' = B \otimes_{A,F_A} A$ and consider the ring map $F_{B/A} : B' \to B$, $b \otimes a \mapsto b^p a$. Then our assertion is that $\Omega_{B/A} = \Omega_{B/B'}$. This is true because $\mathrm{d}(b^p a) = 0$ if $\mathrm{d} : B \to \Omega_{B/A}$ is the universal derivation and hence d is a B'-derivation. □

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